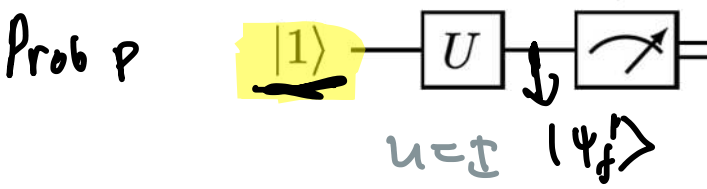


State preparation noise

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Density operator

$$= P_{\text{no error}} P_{\text{ideal}} + P_{\text{error}} P_{\text{error}}$$

$$P_f = P_{noerr} P_{ideal}^{final} + P_{err} P_{err}^{final}$$

$$(U|\psi\rangle)^\dagger = (|\psi\rangle)^\dagger U^\dagger$$

$$\langle\psi_f| = \langle\psi|U^\dagger$$

$$\rho_v = u_p u^+$$

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

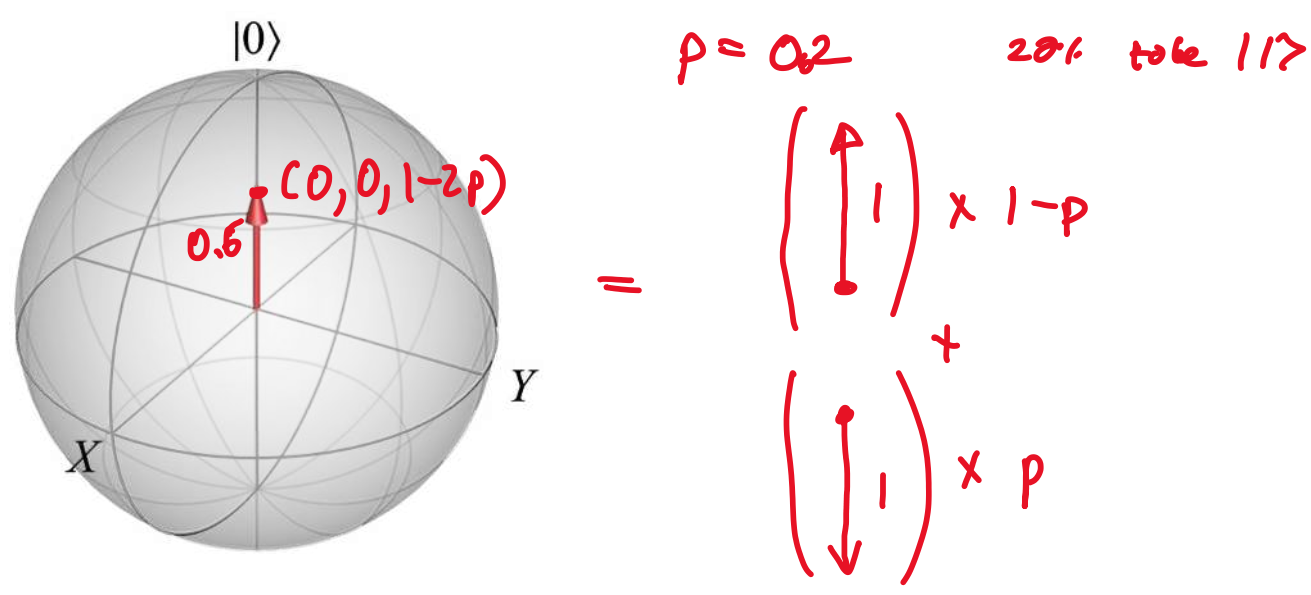
$$||\gamma|| = \begin{pmatrix} 0 & 6 \\ 0 & 1 \end{pmatrix}$$

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & \langle \hat{Z} \rangle \\ 0 & 1 \end{pmatrix} = \frac{1}{2} (\hat{I} + \langle \hat{Z} \rangle \hat{Z})$$

$$\langle X \rangle = \text{Tr}[X\rho] = 0$$

$$\langle Y \rangle = \text{Tr}[Y \rho] = 0$$

$$\text{Tr}[P_\alpha P_\beta] = 2\delta_{\alpha\beta} \quad \text{for } \alpha, \beta \in \{x, y, z\}$$



$$\text{Tr}(p^2) = \langle X \rangle^2 + \langle Y \rangle^2 + \langle Z \rangle^2 = 0^2 + 0^2 + (1-2p)^2$$

Graph of $\text{Tr}(p^2)$ versus p . The curve is a parabola opening upwards, with its minimum at $p = 0.5$. The y-axis is labeled $\text{Tr}(p^2)$ and has a tick mark at 1. The x-axis is labeled p and has tick marks at 0 and 1. The curve starts at $(0, 1)$, goes down to $(0.5, 0)$, and goes back up to $(1, 1)$. Red handwritten notes are present: $1 > 0.5$ near the start, 0.5 near the minimum, and $1 > 0.5$ near the end.

Scaling to larger number of qubits

$$[(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|]^{\otimes n} \equiv U \equiv \text{[Measurement]}$$

$$\rho_0 = \left[(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1| \right]^{\otimes n}$$

$$= (1-p)^n |00\dots 0\rangle\langle 0\dots 0| + \dots + |10\dots 0\rangle\langle 1\dots 1| + \dots$$

$$P(X = 0000 \dots 0) = \begin{cases} 1 & \text{if } p=0 \text{ (ideal case)} \\ (1-p)^n \approx 1 - np + O(p^2) & \text{for } p>0 \end{cases}$$

$$\langle Z_k \rangle = \langle ||| \overset{\vee}{Z} ||| \dots \rangle = 1 - 2p \quad \xrightarrow{\text{much faster}}$$

$$\langle z z z \dots z \rangle = \text{Tr} [z^{\otimes n} \rho] = (1 - 2p)^n$$

$$\rho_0 = \prod_{k=1}^n \frac{1}{2} (\hat{I}_k + (1-2p) \hat{Z}_k)$$

$$= \frac{1}{2^n} (1 + \underline{-2})(1 + \underline{-2}) \dots$$

$$= \frac{1}{2^n} \int (1-2p)^n z^{\otimes n} + \dots$$

$$T_{\mathbb{C}}\{P^2\} = \prod_{i=1}^n T_{\mathbb{C}}(P^2) = (1 - z_i)^{2n}$$

